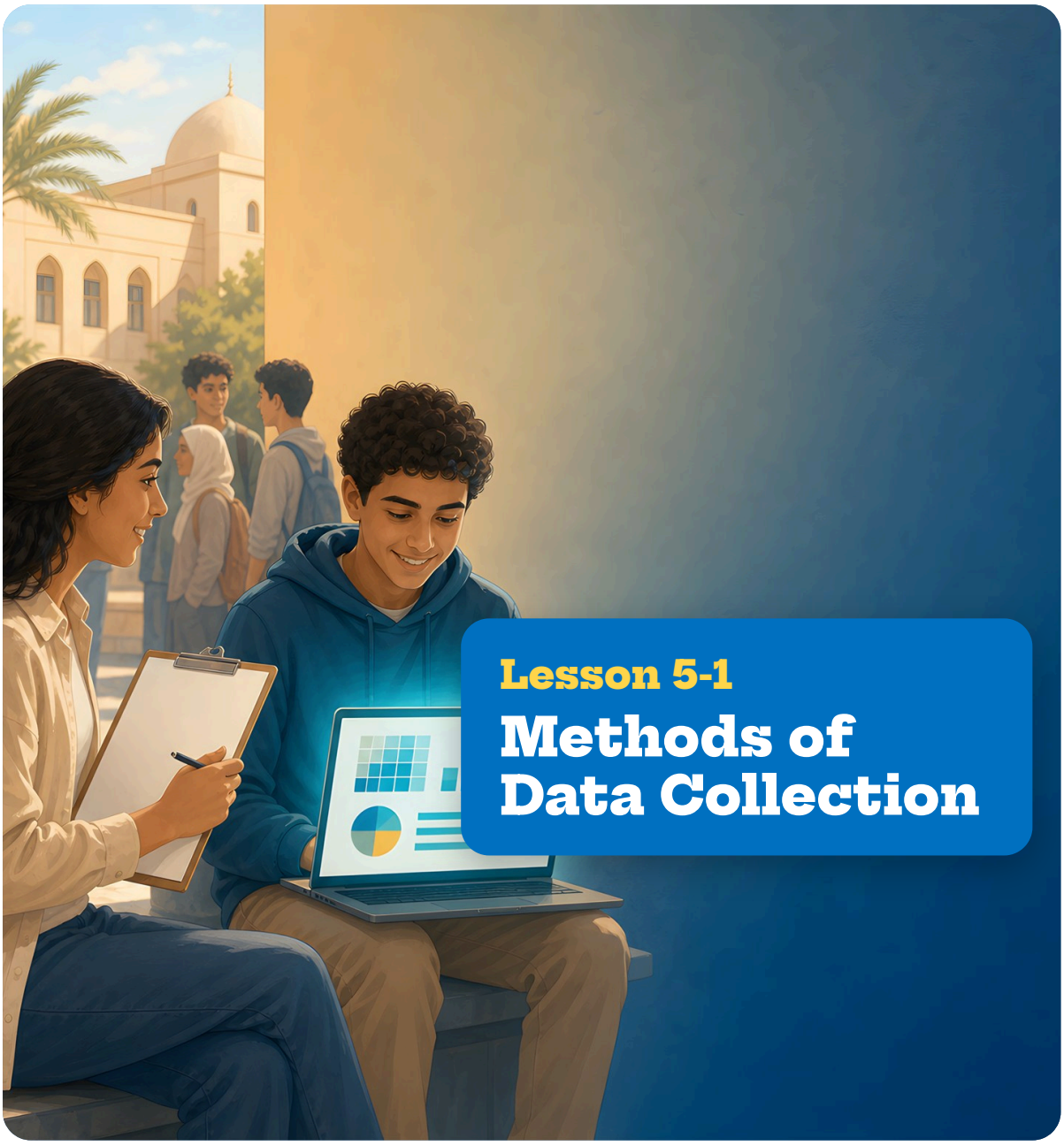


Learning Objectives

- Distinguish primary from secondary data and explain when each is appropriate.
- Explain how sampling works, and describe how sampling bias can lead to confident but wrong conclusions.

Before you can trust what data tells you, you have to trust how it was gathered. Good analysis moves through three stages — collection, organization, then analysis — and if the wrong data is collected, no careful work later can rescue it. Some data you gather yourself for your exact purpose (primary data); some already exists, collected by others (secondary data). And when a whole group is too large to survey, you study a sample — but only a fair sample lets you speak for the whole. A biased sample gives a confident answer that is simply wrong.

? Guiding question: How do you collect data you can trust, and how can a small sample fairly represent a whole population?

Point!

1. The Data Collection Stage

(1) Data analysis progresses through the flow of "collection → organization → analysis," after the purpose has been clarified. In **(data collection)**, the first stage, it is necessary to plan what data will be gathered and how, in line with the purpose of the analysis.

(2) If data that does not match the purpose is collected, it is impossible to draw appropriate conclusions, no matter how carefully later organization and analysis are carried out. For this reason, data collection is an important stage that determines the quality of the entire analysis.

2. Primary Data and Secondary Data

(1) **(Primary data)**…data that is newly collected to fit a specific purpose. *e.g.,* surveys, interviews, observations, experiments. Strength: the items collected can be designed to match the purpose, providing excellent originality and reliability. Constraint: collection takes time, effort, and cost.

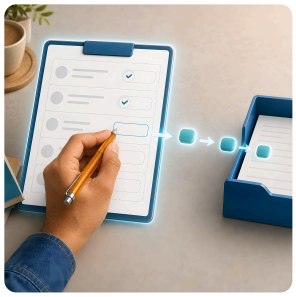
(2) **(Secondary data)**…data that has already been collected and made public by others, and is used in analysis. *e.g.,* open datasets, existing statistics, industry reports. Strength: large amounts of data can be obtained in a short time. Constraints: it may not perfectly match one's purpose; bias or omissions may have arisen during editing or summarizing.

(3) Primary data, since it can be collected to match one's own purpose, has excellent reliability and originality, making it a high-value type of data that can serve as the core of analysis. On the other hand, the value of secondary data lies in being able to easily make use of large-scale data and existing statistics that cannot be obtained from primary data alone. In actual analysis, the two are appropriately combined depending on the purpose and constraints.

Key Fact:
Good analysis starts with good collection: primary data is gathered fresh for a purpose; secondary data already exists; sampling studies part of a population to estimate the whole — and only an unbiased sample supports valid conclusions.

Key Concepts:

- data collection
- primary / secondary data
- population & sample
- sampling (simple random, stratified)
- sampling bias



Sound collection is the base of trustworthy data.

KEY TERMS
data collection — the first stage: plan what to gather and how.
primary data — collected fresh for a purpose.
secondary data — already collected by others.

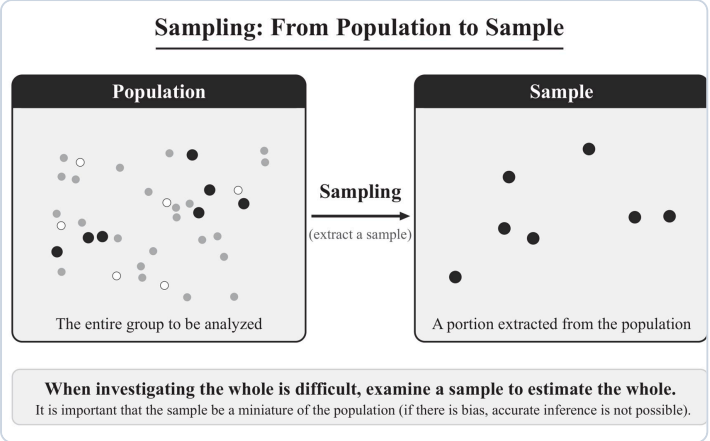


Primary data is gathered fresh; secondary is reused.

3. Sampling

(1) The entire group that is the target of analysis is called the **(population)**, and the part of the population that is actually surveyed is called the **(sample)**. Surveying the entire population is called a **(complete survey)**, also called a census, and surveying a chosen sample is called a **(sample survey)**.

(2) **(Sampling)**…selecting a sample from the population. Depending on the sampling method, the reliability of the conclusions obtained varies greatly.



Sampling: from population to sample

(3) A typical sampling technique is **(random sampling)**, as described below. Random sampling makes it likely that the sample represents the population, allowing the analysis results to be applied to the population as a whole.

Method	Description
(Simple random sampling)	Choose a sample from the population purely by chance, without bias
(Stratified sampling)	Divide the population into strata such as gender or age group, and randomly choose from each stratum

(4) Simple random sampling is the most basic method and can be used even when little is known about the population. Stratified sampling can reflect the composition of the population (such as the ratios of gender and age group) in the sample, making it easier to reduce bias when differences between strata affect the analysis.

4. Bias in Surveys

(1) **(Sampling bias)**…this occurs when the method of selecting a sample is biased and does not correctly reflect the actual state of the population. When sampling bias is present, the conclusions obtained cannot be applied to the population as a whole.

(2) Typical examples of bias. Self-selection and time/location bias concern who is surveyed (selection bias); response bias concerns how those surveyed answer (measurement error).

PAUSE & THINK

A government census you download — is that primary or secondary data for you?



A fair sample mirrors the whole; a biased one misleads.

KEY TERMS

population — the whole group analyzed.
sample — the part actually surveyed.
sampling bias — an unfair sample that misrepresents the whole.

Type of bias	Description	Example
(Self-selection bias)	Bias caused by only those who want to respond actually responding	In a web survey, only people with high interest respond
(Response bias)	Not a selection problem: people already surveyed give answers that look desirable rather than accurate	Reporting a “study time” longer than it really was
(Time/location bias)	Bias caused by the timing or location of the survey being limited	Surveying in front of a station on weekday afternoons, missing office workers and students

(3) It is difficult to eliminate sampling bias completely, but bias can be reduced by carefully designing the survey: how subjects are chosen, how questions are worded, and how the survey is conducted.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Primary data refers to data that has already been collected and made public by others.

B Selecting a sample from a population is called sampling.

C Stratified sampling is a method in which the population is divided into strata, and a sample is randomly chosen from each stratum.

D When sampling bias is present, the conclusions obtained may not be applicable to the population as a whole.

(2) Match each description (a, b) of a sampling method with the most appropriate option from the choices below (A, B).

a Choose a sample from the population by chance, without bias.

b Divide the population into strata such as gender or age group, and randomly choose from each stratum.

[Choices] **A** Stratified sampling **B** Simple random sampling

✓ Solution

(1)

A Primary data is data that is newly collected to fit a specific purpose. Data already collected and made public by others is secondary data. Therefore, ×.

B Correct definition of sampling. Therefore, ○.

C Correct definition of stratified sampling. Therefore, ○.

D Correct description of the effect of sampling bias. Therefore, ○.

💡 PAUSE & THINK

A web poll anyone can enter — whose voices are missing, and which bias is that?

(2)

a: Choosing without bias by chance is simple random sampling. **B**

b: Dividing into strata and choosing randomly from each is stratified sampling. **A**

Try

1 Answer the following questions.

1. What is the term for data that is newly collected to fit a specific purpose?
2. What is the term for data that has already been collected and made public by others, and is used in analysis?
3. What is the term for selecting a sample from a population?
4. What is the term for when the method of selecting a sample is biased and does not correctly reflect the actual state of the population?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is most appropriate as a method of collecting primary data.

A Use statistical data published by a government agency.

B Create a survey to fit one's own purpose and collect responses.

C Cite figures from a report issued by an industry association.

D Use survey data from a study conducted earlier by someone else as is.

(2) For each survey situation (a, b), choose the most appropriate sampling method from the choices below (A, B).

a When investigating study time across all students in a school, you want to choose the sample taking into account differences in tendencies between grades.

b You want to grasp the opinions of all eligible voters in a city without bias.

[Choices] **A** Simple random sampling **B** Stratified sampling

EXAM-STYLE QUESTION

Explain why a biased sample can make an analysis worthless even when the calculations are correct. **[6] Refer to:** the sample as a miniature of the population.

PAUSE & THINK

Which option gathers new data *you* designed — and which just reuses others' data?

(3) From the following options (A–D), choose the one that is most appropriate as an example in which sampling bias has occurred.

A Surveying 100 students randomly chosen from all students in the school by drawing lots.

B Dividing all residents into age-group strata and surveying a randomly chosen sample from each stratum.

C Approaching passers-by in front of the station on weekday afternoons and reporting the results as “the opinions of all citizens.”

D Designing the wording of a survey question carefully so that respondents can answer truthfully.

Think as an Engineer — Investigate & Decide

Your school wants to know how long students typically spend commuting to school each day, but you cannot survey everyone. You design how to choose and question a sample.

☐ **Ask whoever is easiest to reach**

☐ **Plan a fair, representative sample**

1 · Collect data (plan the sample). Decide how many students you would ask and how you would pick them so the sample fairly represents the whole school. As a quick pilot, tally the commute of 8–10 classmates. State the population and the sample clearly.

2 · Analyze the risk of bias. Name one group your method might silently drop — for example, students who were absent that day, or students who never open an online survey link. Explain how missing that group would skew the estimated commute, and say which bias it is closest to (self-selection, response, or time/location).

3 · Decide. Choose simple random or stratified sampling for this survey and justify the choice from the risk you found in step 2 — think about whether commute really differs between identifiable subgroups here. **In pairs,** compare methods and agree on the single change that would most reduce bias.

Give evidence for your answer.

Stuck? Start from who might be left out. Ask yourself: could one part of the school answer very differently from the rest — a whole group, or students who never see your survey? If a subgroup might differ systematically, the real question is how to give every subgroup a fair chance *before* you pick your method. Work out which subgroups matter here, then let that guide your choice.

🌍 In a New Context

A shopping mall surveys only visitors on Saturday afternoon and reports the results as “what all our customers think.” Name the bias, explain who is missed, and describe one change that would make the sample fairer.

❓ Lesson Question — Answered

The lesson asked how to collect data you can trust and how a small sample can represent a whole population. Trustworthy analysis begins with collection: after the purpose is clear, you choose primary data (gathered fresh for that purpose, with high reliability but more cost) or secondary data (already collected by others, fast to obtain but perhaps mismatched or biased), often combining both. When the population — the whole group analyzed — is too large, you study a sample through sampling, aiming for a miniature of the population. Random sampling makes that likely: simple random sampling picks purely by chance, while stratified sampling splits the population into strata and picks from each to mirror its make-up. The danger is sampling bias, where a skewed selection cannot speak for the whole; it is reduced by careful design of who is chosen, how questions are worded, and how the survey is run.

KEY TERMS

simple random sampling — picked purely by chance.
stratified sampling — split into strata, then pick from each.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

In data analysis, it is important to plan what data to collect according to the purpose. The data to be collected can be classified into two types: (a), which is newly collected, and (b), which has already been collected and made public by others. Also, when investigating the entire population is difficult, (c) is performed to select a part of the population. At this time, if bias arises in the way of selection, (d) occurs, and the analysis results may not be applicable to the population as a whole.

1. Fill in the blanks (a)–(d).

⭐ REFLECT & CHALLENGE

Reflect: Think of an online poll you have seen. Whose voices were probably missing from it, and why? **Challenge:** You may survey only 40 of your school's 800 students about a new schedule. Describe exactly how you would choose those 40 to avoid bias, and name the one group you would make sure is represented.

3 Answer the following questions.

(1) From the following options (A–D), choose the one that is most appropriate as an example of secondary data.

A Survey responses collected from students in one's own class.

B Population statistics published by the government.

C Traffic on a school commute route observed and recorded by oneself.

D Records of an interview designed and conducted by oneself.

(2) From the following options (A–D), choose the one that most appropriately describes sampling methods.

A Simple random sampling is a method in which the surveyor preferentially chooses subjects that are easy to gather.

B Stratified sampling is a method in which the population is divided into strata, and a sample is randomly chosen from each stratum.

C Simple random sampling is a method in which the population is divided into strata, and a representative is chosen from each stratum.

D A complete survey is a method in which only part of the population is surveyed.

(3) From the following options (A–D), choose the one that is NOT appropriate as a way to reduce sampling bias.

A Choose subjects without bias from the entire population.

B Limit the timing and location of the survey, and gather responses only from people in specific conditions.

C Choose the sample taking into account the strata of the population (age group, gender, etc.).

D Design the wording of survey questions so that respondents can answer truthfully.

⭐ Key takeaway

Remember:
Collect the right data (primary vs secondary), study a fair sample of the population, use random sampling (simple or stratified) so the sample mirrors the whole, and guard against sampling bias.